Endogenous Networks and Inflation

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Motivation

Supply chain disruptions have been increasingly frequent in the recent years, arising from pandemics, geopolitical tensions, trade policy, natural disasters and extreme weather events, cybersecurity threats and transportation bottlenecks, among other factors. Empirical evidence shows that such disruptions lower firms' output and raise their marginal costs, putting upward pressure on inflation.

The recent literature on macroeconomic fragmentation emphasizes an efficiency–security trade-off: reliance on concentrated foreign sourcing raises production efficiency but increases exposure to supplier-specific risk factors, whereas diversification, de-risking and reshoring improve resilience at the cost of a lower output and higher marginal cost.[1]

Even if initially costlier than sourcing all inputs from the most productive supplier, endogenous investment in supply-chain resilience can attenuate the price response to supply disruptions. One could hypothesize that a worsening of the security–efficiency trade-off (for example, from an increase in uncertainty about the productivity of, or the access to critical foreign suppliers) can induce costly reallocation toward more resilient sourcing structures, which, in turn, dampen the inflationary effect of a material disruption.

Research question

Therefore, the central research question I want to answer is: **does an increase in uncertainty about the productivity of, or access to critical suppliers lower the pass-through of subsequent supply disruptions to inflation?**

The goal of my thesis is to write a theoretical model that can provide an answer to this question. The results of such a model would inform on whether the inflation generated by uncertainty shocks is a pure cost or is partly the price of a less risky network.

This model would also be relevant to monetary policy. Rubbo (2023) shows that, in a multi-sector economy with input-output linkages, sectoral productivity shocks break the divine coincidence by entering the Phillips curve as an endogenous cost-push term, whose size depends on the input-output matrix and on price and wage rigidities. Stabilizing consumer-price inflation then forces the central bank to accept an output gap. In a model where the input-output matrix responds to supplier risk, the severity of this trade-off becomes an equilibrium outcome. This leads to a second question: **does the network adaptation shrink the cost-push term generated by a subsequent supply chain disruption?**

[1] Ahn and Tan 2025; Grossman, Helpman, and Lhuillier 2023; Grossman, Helpman, and Sabal 2024; Castro-Vincenzi et al. 2024.

Building blocks

A model that can account for the effect of uncertainty shocks on inflation and characterize the conditions under which the supply chains adaptation immunizes inflation against future shocks needs two main blocks: 1) a network block that models endogenous, risk-driven sourcing; 2) a New Keynesian overarching setting in which inflation, rather than relative prices, can be analyzed.

Kopytov et al. (2024) study endogenous network formation in a setting where firms choose an input mix before productivity shocks are realized. Firms therefore weigh a supplier's expected productivity draw against its volatility. When uncertainty rises, they shift toward more stable suppliers even at higher prices. Aggregate volatility falls, but so does output.

Rubbo (2023) revisits the New Keynesian framework in a multi-sector economy with input-output linkages and derives analytical expressions for sectoral and aggregate Phillips curves (and their cost-push term) and welfare, as a function of (network) primitives.

Kopytov et al. (2024) let the network respond to risk, but their economy has flexible prices and no monetary policy, so a shift toward safer suppliers shows up as lower output and lower volatility, never as inflation. Rubbo (2023) delivers inflation, but takes the input-output matrix as given.

My plan is to combine the two. I take the network block of Kopytov et al. (2024) as the mechanism through which firms choose their input mix under uncertainty and embed it in the multi-sector New Keynesian setting of Rubbo (2023). The input-output matrix that is a calibrated parameter in the latter becomes an equilibrium object responding to suppliers' risk.

To answer the two research questions, the model must deliver two objects as functions of the network that firms choose under uncertainty. The first is the response of CPI inflation to a supply disruption, under a given monetary policy rule. The second is the cost-push term that the disruption generates in the CPI inflation Phillips curve. Whether adaptation raises or lowers the response of inflation and the cost-push is an open question and conditions determining the direction and magnitude of this effect is the purpose of the exercise.

References

[1] JaeBin Ahn and Brandon Tan. "Supply Chain Diversification and Resilience". In: *IMF Working Papers* 2025.102 (May 2025), p. 1. DOI: https://doi.org/10.5089/9798229011419.001.

[2] Juanma Castro-Vincenzi et al. *Weathering the Storm: Supply Chains and Climate Risk*. Working Paper 32218. National Bureau of Economic Research, Mar. 2024. DOI: https://doi.org/10.3386/w32218.

[3] Gene M. Grossman, Elhanan Helpman, and Hugo Lhuillier. "Supply Chain Resilience: Should Policy Promote International Diversification or Reshoring?" In: *Journal of Political Economy* 131.12 (2023), pp. 3462–3496. DOI: https://doi.org/10.1086/725173.

[4] Gene M. Grossman, Elhanan Helpman, and Alejandro Sabal. "Optimal Resilience in Multi-tier Supply Chains". In: *The Quarterly Journal of Economics* 139.4 (Nov. 2024), pp. 2377–2425. DOI: https://doi.org/10.1093/qje/qjae024.

[5] Alexandr Kopytov et al. "Endogenous Production Networks Under Supply Chain Uncertainty". In: *Econometrica* 92.5 (2024), pp. 1621–1659. DOI: https://doi.org/10.3982/ECTA20629.

[6] Elisa Rubbo. "Networks, Phillips Curves, and Monetary Policy". In: *Econometrica* 91.4 (2023), pp. 1417–1455. DOI: https://doi.org/10.3982/ECTA18654.